

Week-1 Assignment

Ques 1. Portfolio Loss Dependence

sol.
Given : $P(LA) = 0.10$, $P(LB) = 0.20$, $P(LA|LB) = 0.35$.
Indicators : $IA = \begin{cases} 1 & \text{if A loses} \\ 0 & \text{otherwise} \end{cases}$ $IB = \begin{cases} 1, & \text{if B loses} \\ 0, & \text{otherwise} \end{cases}$

and $N = IA + IB$

(a).

$$P(LA \cap LB) = P(LA|LB) P(LB) \\ = 0.35 \times 0.20 = 0.07$$

$$P(LA \cup LB) = P(LA) + P(LB) - P(LA \cap LB) \\ = 0.10 + 0.20 - 0.07 = 0.23$$

(b) $P(LA)P(LB) = (0.10)(0.20) \\ = 0.02 \\ \neq 0.07 (P(LA \cap LB))$

$\therefore$ Not independent

Also, $P(LA|LB) = 0.35$

but $P(LA) = 0.10$

Not equal $\Rightarrow$ dependence exists
 $\therefore LA, LB$ are dependent

(c) $N = IA + IB$

Expected value : $E(N) = E(IA) + E(IB)$

$$E(IA) = P(LA) = 0.10$$

$$E(IB) = P(LB) = 0.20$$

$$E(N) = 0.10 + 0.20$$

$$E(N) = 0.30$$

variance

$$\text{var}(N) = \text{var}(IA) + \text{var}(IB) + 2 \text{cov}(IA, IB)$$

$$\text{var}(IA) = 0.10 \times 0.90 = 0.09$$

$$\text{var}(IB) = 0.20 \times 0.80 = 0.16$$

covariance

$$\text{cov} = 0.07 - (0.10)(0.20) = 0.05$$

Hence

$$= 0.09 + 0.16 + 2(0.05) = 0.35$$

(d). If independence is assumed:

$$P(LA \cap LB) = P(LA)P(LB) = (0.10)(0.20) = 0.02$$

$$\text{Diff} = 0.07 - 0.02 = 0.05$$

$$\text{underestimation \%} = \frac{0.05}{0.07} \times 100 = 71.43 \%$$

$$\therefore \text{Naive estimate} = 0.02$$

$$\therefore \text{True value} = 0.07$$

$$\therefore \text{underestimate joint risk by: } 0.05$$

(e) dependence affect portfolio risk because loss may occur simultaneously

- Portfolio risk
- value at risk
- expected losses
- stress scenario

② Markov Chain Migration

(a) Directed graph

Transition:

State 0 $\rightarrow$ 0, 1, 2, 3

State 1 $\rightarrow$ 1, 2

State 2 $\rightarrow$ 1, 2

State 3 $\rightarrow$ 3

Graph: $0 \rightarrow \{0, 1, 2, 3\}$
 $1 \leftrightarrow 2$
 $3 \rightarrow 3$

(b) Communicating classes: $\{0\}, \{1, 2\}, \{3\}$

Closed classes:

- $\{1, 2\} \rightarrow$ closed
- $\{3\} \rightarrow$ closed
- $\{0\} \rightarrow$ not closed

(c) Closed classes $\rightarrow$ recurrent

Thus, recurrent: $\{1, 2, 3\}$

transient: $\{0\}$

(d) No, because all states do not communicate

class $\{1, 2\} \rightarrow$ self loop exists $\rightarrow$ aperiodic

class $\{3\} \rightarrow$ absorbing $\rightarrow$ aperiodic

(e) class $\{3\}$

absorbing

state: $\pi^{(3)} = (0, 0, 0, 1)$

class $\{1, 2\}$

$P_{\{1, 2\}} = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$

We get $\pi_1 = \pi_2 = 0.5$

extending to full space: $\pi^{(1, 2)} = (0, 0.5, 0.5, 0)$

(b) Take convex combinations:

$$\pi = \alpha(0, 0.5, 0.5, 0) + (1-\alpha)(0, 0, 0, 1)$$

where $0 \leq \alpha \leq 1$

thus, $\pi = (0, \frac{\alpha}{2}, \frac{\alpha}{2}, 1-\alpha)$

(g), (h) $\rightarrow$ code

(i) credit risk interpreter

- class (3) = Default state
- class (1, 2) = Persistent risky states
- ~~class~~ state 0 is transient because investments -
grade bonds may deteriorate and leave permanently

(3) Inventory random walks

(a) recurrent origin - Inventory returns to neutral position (0) infinitely often

Transient origin - inventory may leave neutrality and eventually never returns

(b) state i is recurrent if $\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$

Transient if $\sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty$

(c) when $p_{00}^{(2n)} \sim \frac{C}{n^{d/2}}$

$\bullet d=1: \sum \frac{1}{\sqrt{n}}$

diverges $\Rightarrow$ recurrent

$\bullet d=2: \sum \frac{1}{n}$

diverges $\Rightarrow$ recurrent

$d=3: \sum \frac{1}{n^{3/2}}$

converges $\Rightarrow$ transient

(d) code

(e) - code

(f) path length - $n = 100, 1000, 10000$

Obs

- $\uparrow$ path length $\Rightarrow$ returns probab $\uparrow$
- $d = 1, 2$: approaches 1 (recurrence)
- $d = 3$: $\uparrow$ slowly but stays below 1.

(g). simulation obscure only:

- finite paths
- finite time horizon

Recurrence / transience depends on: $n \rightarrow \infty$

Hence, simulation can suggest behaviour but cannot mathematically prove it

(ii) Market Regime Switching

Sol.

(a)

- All state commⁿ (can reach each other eventually)

$\Downarrow$

irreducible

- self loop exist $P_{00}, P_{11}, P_{22}, P_{33} > 0$

Hence aperiodic

(b) solving $\pi P = \pi$

$$\text{and } \pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$$

Result $\times$

$$\pi = (0.269, 0.327, 0.231, 0.173)$$

(c) π_i = long run fraction of weeks spent in regime i

thus, Bull = 26.9%

Neutral = 32.7%

Bear = 23.1%

crisis = 17.3%

⑥

$$(d) \bar{r} = \sum_{i=0}^3 \pi_i r_i$$

$$= (0.269)(0.012) + (0.327)(0.004) + (0.232)(-0.008) + (0.173)(-0.025)$$

$$\bar{r} \approx -0.0012$$

$\therefore -0.12\%$ per week

(e) Bear + Crisis ~~is~~

$$\pi_2 + \pi_3$$

$$= 0.232 + 0.173$$

$$= 0.404$$

$$\approx 40.4\%$$

(f) Python code

(g) Python code

(h), (i)

(j) simulation helps -

- validate theoretical results
- observe simple paths
- estimate quantities difficult analytically
- visualize behaviour